Overview of Data Warehousing

Part 4 to the End (Slides 14–24)

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1 Why OLAP and OLTP Don't Mix

OLTP and OLAP are often separated because they have **different performance requirements**, **different data modeling requirements**, and **different data sources**.

1.1 Different Performance Requirements

OLTP:

- Fast response time is important (less than 1 second in the lecture's example).
- Data must be up-to-date and consistent at all times.

OLAP:

- Complex queries may consume many resources.
- Queries can saturate CPU and disk bandwidth.
- A static snapshot of data is usually acceptable.

1.2 Different Data Modeling Requirements

OLTP:

- Uses **normalized schemas** for consistency.
- Often has complex data models and many tables.
- Uses a limited number of standardized queries and updates.

OLAP:

- Simplicity of the data model is important.
- **Denormalized schemas** are common.
- Fewer joins can improve query performance.
- Fewer tables make the schema easier to understand.

1.3 Analysis Requires Many Data Sources

- OLTP usually focuses on one specific process (e.g., online orders).
- OLAP integrates data from different processes (e.g., sales, inventory, and purchasing).
- OLAP often uses **historical data** to find long-term patterns and changes in behavior.
- Different terminology and schemas make data integration challenging.

2 Data Warehouse as a Solution

Using OLTP and OLAP in the same database system is often impractical. The solution is to build a **Data Warehouse**.

1. Copy data from various OLTP systems.
2. Optimize data organization and system tuning for OLAP.
3. Prevent large analysis queries from slowing daily transactions.
4. Periodically refresh the warehouse data.

3 The Warehousing Approach

In this approach, information is **integrated in advance** and stored in a warehouse for direct querying and analysis.

3.1 Main Architecture Components

1. **Sources:** Original systems containing the data.
2. **Extractors/Monitors:** Extract data and monitor the sources.
3. **Integration System:** Integrates data from different sources.
4. **Metadata:** Descriptive information used by the integration system.
5. **Data Warehouse:** Stores integrated information.
6. **Clients:** Query and analyze the information.

Basic flow: Sources → Extractors/Monitors → Integration System → Data Warehouse → Clients.

3.2 Advantages

1. **High query performance**, but the information is not necessarily the most current.
2. Complex queries at the warehouse do not interfere with local OLTP processing.
3. Copied information can be modified, annotated, summarized, restructured, and stored historically.

4 Need for Data Warehousing

Companies have gathered huge volumes of data over the years. Data warehousing helps them analyze this data and use it to gain a **competitive advantage**.

5 Benefits of Data Warehousing

1. More efficient analysis of data.
2. Competitive advantage.
3. Support for strategic decision making.
4. Increased productivity of decision makers.

6 Decision Support Systems (DSS), DW and OLAP

DSS (Decision Support System): Information technology that helps executives, managers, and analysts make faster and better decisions.

- DSS provides current and historical information for decision support.
- The lecture describes a **Data Warehouse** as a Decision Support System.
- **OLAP** is an element of DSS used for analytical processing.

7 Data Warehouse Characteristics

1. Analysis-driven.
2. Supports ad-hoc queries.
3. Supports complex queries.
4. Used by top managers.
5. Based on dimensional modeling.
6. Uses denormalized structures.

8 Data Warehousing Applications

- **Retail:** Customer loyalty and market planning.
- **Financial:** Risk management and fraud detection.
- **Airlines:** Route profitability and yield management.
- **Manufacturing:** Cost reduction and logistics management.
- **Utilities:** Asset management and resource management.
- **Government:** Manpower planning and cost control.

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